

Behavioural Cost and COVID-Safe Behaviour Adherence Over Time in Australia

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How did COVID-safe behaviour adherence change over time in Australia?

This project asks:

- Which behaviours had higher adherence?
- Did easier behaviours stay more stable?
- What explains the 2021 surge?

Main idea

Adherence depends on both behavioural burden and outbreak context.

Data and Approach

Data

- Imperial College London / YouGov COVID-19 Behaviour Tracker
- Australia subset
- 53,833 responses
- 54 survey waves

Approach

- 14 protective behaviours
- Scores 4–5 coded as adherent
- Weighted adherence percentages
- Trends summarised using 6-week blocks
- States compared during surge period

How to interpret the trends

Each survey wave includes different respondents, so the trends show population-level changes over time rather than changes within the same individuals.

What is Behavioural Cost?

Simple definition

Behavioural cost means how difficult a protective behaviour is to follow in everyday life. A higher cost score means the behaviour is more burdensome to maintain.

Each behaviour was scored using five burden dimensions:

- Practical effort
- Social disruption
- Lifestyle restriction
- Emotional burden
- Financial burden

Scoring and classification

Each dimension was scored from 1–3. The five scores were added to classify behaviours into low, medium, and high behavioural cost groups.

Behavioural Cost Groups

How the groups were formed

Each behaviour was given a literature-informed burden score across five dimensions. Total scores could range from 5–15: 5–6 = low burden, 7–9 = moderate burden, and 10–15 = high burden.

Low cost

- cough/sneeze covering
- handwashing
- sanitiser use

Medium cost

- mask wearing
- avoid crowds
- avoid guests
- avoid public transport
- avoid symptomatic people

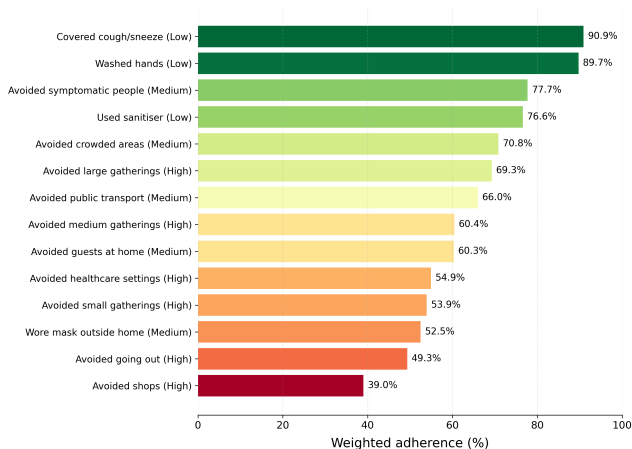
High cost

- avoid gatherings (small, medium, large)
- avoid going out
- avoid shops
- avoid healthcare

Main purpose

This grouping tests whether lower-burden behaviours remained more stable over time than higher-burden behaviours.

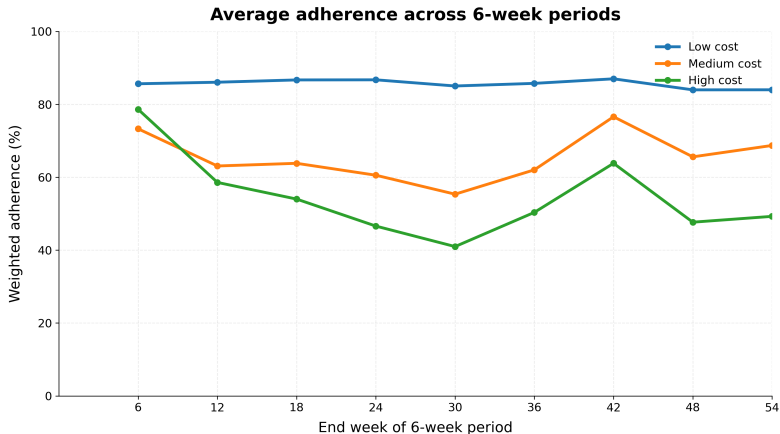
Overall Adherence by Behaviour



Key message

Hygiene behaviours had high adherence, but behavioural cost did not explain all differences.

Adherence Trends Over Time



Key message

Low-cost behaviours stayed stable, while medium- and high-cost behaviours followed similar fluctuating patterns.

Main Surge Period

The biggest change happened between two 6-week periods.

Before surge

Weeks 31–36

5 May–27 July 2021

After surge

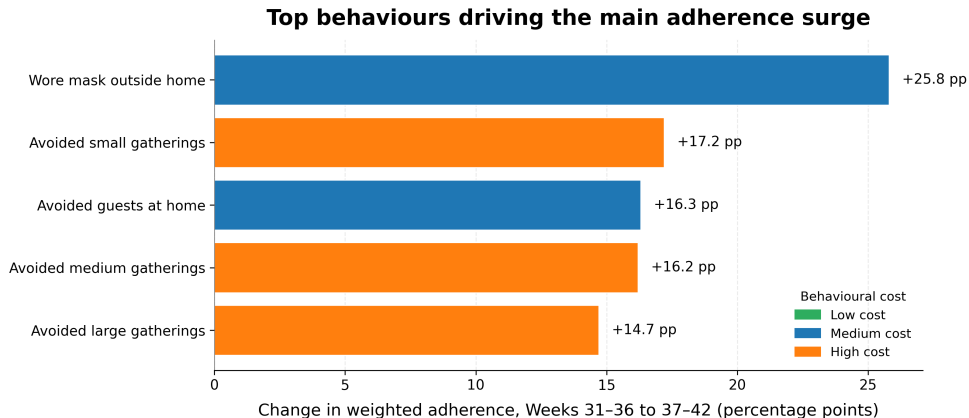
Weeks 37–42

28 July–17 October 2021

Interpretation

This overlaps with the 2021 Delta outbreak and stronger state restrictions.

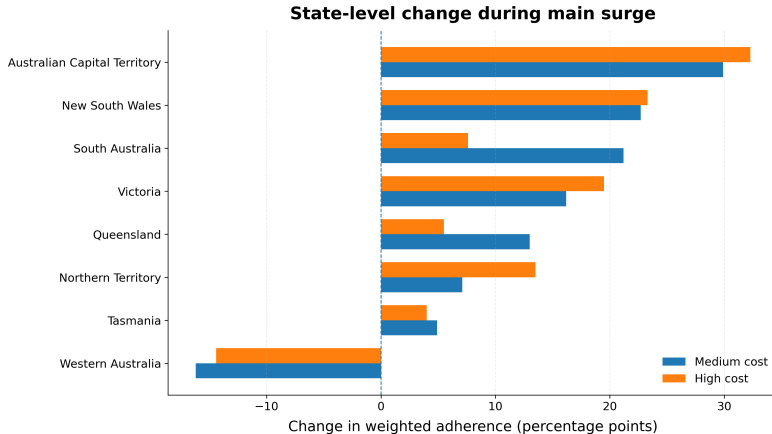
Behaviours Driving the Surge



Key message

Mask wearing increased the most, followed by gathering and visitor avoidance behaviours.

State-Level Surge



Key message

The surge was not equal across Australia. State policy context matters.

Why Did This Surge Happen?

The timing matches the 2021 Delta period.

- stronger mask rules
- lockdowns in key states
- limits on guests and gatherings
- higher public risk perception
- stronger policy and media attention

Careful interpretation

This does not prove causation, but the timing supports the explanation.

Main Takeaway

COVID-safe adherence did not change equally across behaviours.

- hygiene behaviours stayed high and stable
- mask wearing increased the most
- gathering avoidance also increased strongly
- state context helped explain the surge

Conclusion

Behavioural cost matters, but outbreak and policy context also shaped adherence.

This analysis gives a descriptive view of how COVID-safe adherence changed over time. The next step would be to strengthen the explanation behind these changes.

- Link the main surge period more directly with state-level policy timelines.
- Examine mask wearing and gathering avoidance in more detail, because these behaviours drove much of the 2021 increase.
- Use newspaper and media sources to better explain public messaging and perceived risk during the Delta period.
- Compare these descriptive findings with related Australian work on mask mandates and protective behaviour adherence, such as Ryan et al. (2025).

References

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